Learning rate experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete controlled-rate evidence record.

Trace paths,repair confounded comparisons and report training results within their scope.

Keep one habit: W1–W3/W5 → selected new objective → project protocol ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use the same restricted network h = ReLU(x), y = vh, with only v trained. Keep the fictional **training** pairs (1, 2) and (2, 4), start v = 0, and use one full-data batch per update.

  Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

- The **training** objective is ordinary MSE: L(v) = 2.5 times (v - 2) squared. Its gradient is 5 times (v - 2). A step is new v = v - rate times 5 times (v - 2).

  This small quadratic lets us explain each result rather than treating the rate as a magic setting.

- The same rate need not behave this way for another model, **input** scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

Batching and epochs

> **My remembered idea, example and correction**

Updating parameters

> **My remembered idea, example and correction**

Loss functions

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Rate	v after first update	First training MSE	v after four updates
0.05	0.5	5.625	1.3671875
0.2	2	0	2
0.3	3	2.5	1.875
0.4	4	10	0
0.5	5	22.5	-8.125

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main restrictednetwork hReLU(x), onlyvtrained; two pairs(1, 2)/(2, 4), full-dataMSE2.5(v−2)², gradient 5(v−2), start 0, 4 updates. Distance multiplier 1–5 rate:.05→.75 same-side shrinking; .2→0 exact; .3→−.5 alternating shrinking; .4→−1 unchangedloss; .5→−1.5 growing. Mainrate.3v0/3/1.5/2.25/1.875, MSE 10/2.5/.625/.15625/.0390625. Positive0<rate<.4 reducesloss here; not universal.

  Two-row SSEdoublesgradient, so halve rate for same trajectory. Separate picture(a−4)²:rate.25 path2/3 and.75 path6/3 bothloss4/1; rate 1 path8/0 loss16/16. Independent(a−1)²uses a third gradient scale.

- For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main restrictednetwork hReLU(x), onlyvtrained; two pairs(1, 2)/(2, 4), full-dataMSE2.5(v−2)², gradient 5(v−2), start 0, 4 updates. Distance multiplier 1–5 rate:.05→.75 same-side shrinking; .2→0 exact; .3→−.5 alternating shrinking; .4→−1 unchangedloss; .5→−1.5 growing. Mainrate.3v0/3/1.5/2.25/1.875, MSE 10/2.5/.625/.15625/.0390625. Positive0<rate<.4 reducesloss here; not universal.

  Two-row SSEdoublesgradient, so halve rate for same trajectory. Separate picture(a−4)²:rate.25 path2/3 and.75 path6/3 bothloss4/1; rate 1 path8/0 loss16/16. Independent(a−1)²uses a third gradient scale.

- At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main restrictednetwork hReLU(x), onlyvtrained; two pairs(1, 2)/(2, 4), full-dataMSE2.5(v−2)², gradient 5(v−2), start 0, 4 updates. Distance multiplier 1–5 rate:.05→.75 same-side shrinking; .2→0 exact; .3→−.5 alternating shrinking; .4→−1 unchangedloss; .5→−1.5 growing. Mainrate.3v0/3/1.5/2.25/1.875, MSE 10/2.5/.625/.15625/.0390625. Positive0<rate<.4 reducesloss here; not universal.

  Two-row SSEdoublesgradient, so halve rate for same trajectory. Separate picture(a−4)²:rate.25 path2/3 and.75 path6/3 bothloss4/1; rate 1 path8/0 loss16/16. Independent(a−1)²uses a third gradient scale.

- Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main restrictednetwork hReLU(x), onlyvtrained; two pairs(1, 2)/(2, 4), full-dataMSE2.5(v−2)², gradient 5(v−2), start 0, 4 updates. Distance multiplier 1–5 rate:.05→.75 same-side shrinking; .2→0 exact; .3→−.5 alternating shrinking; .4→−1 unchangedloss; .5→−1.5 growing. Mainrate.3v0/3/1.5/2.25/1.875, MSE 10/2.5/.625/.15625/.0390625. Positive0<rate<.4 reducesloss here; not universal.

  Two-row SSEdoublesgradient, so halve rate for same trajectory. Separate picture(a−4)²:rate.25 path2/3 and.75 path6/3 bothloss4/1; rate 1 path8/0 loss16/16. Independent(a−1)²uses a third gradient scale.

- Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

  Propose a repair.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Compare a new objective independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **learning rate** experiments **evidence** sheet to Image-pattern Network Investigation. Show how you can run a **controlled comparison** of **learning rate**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Explain a crossing for **Learning rate** experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Learning rate** experiments. Proposed explanation: "Crossing the minimum means **divergence**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Compare **learning rate**s with identical starting loss and update counts.
- Loss $J(a)=(a-4)^2$; gradient 2(a-4); start a = 0. Compare two updates at rates 0.25 and 0.75.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/14

Learning rate experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Rate 0.25

First a = 2, loss 4; second a = 3, loss 1.

2. Rate 0.75

First a = 6, loss 4; second a = 3, loss 1.

3. Inspect paths

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

CHANGE ONE THING

Add a rate 1 run from a = 0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare **learning rate**s with identical starting loss and update counts.
- Which labelled step matches this detail? Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.
- 1. Inspect paths
- 2. Rate 0.25

- 3. Rate 0.75

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect paths Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Changed-case question and optional hint

- Changed-case practice: Add a rate 1 run from a = 0. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. Its path is 8 then 0, with loss 16 at both steps.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its path is 8 then 0, with loss 16 at both steps. The update alternates across 4 without reducing the initial loss.

Repair and retain the record

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a **learning rate** to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Learning rate experiments: the case

AI ACADEMY / AI-5/14 · 01 · LOOK AT THE FACTS

Learning rate experiments: the case

The situation

Compare learning rates with identical starting loss and update counts.

Supplied evidence

Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain learning rate experiments using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

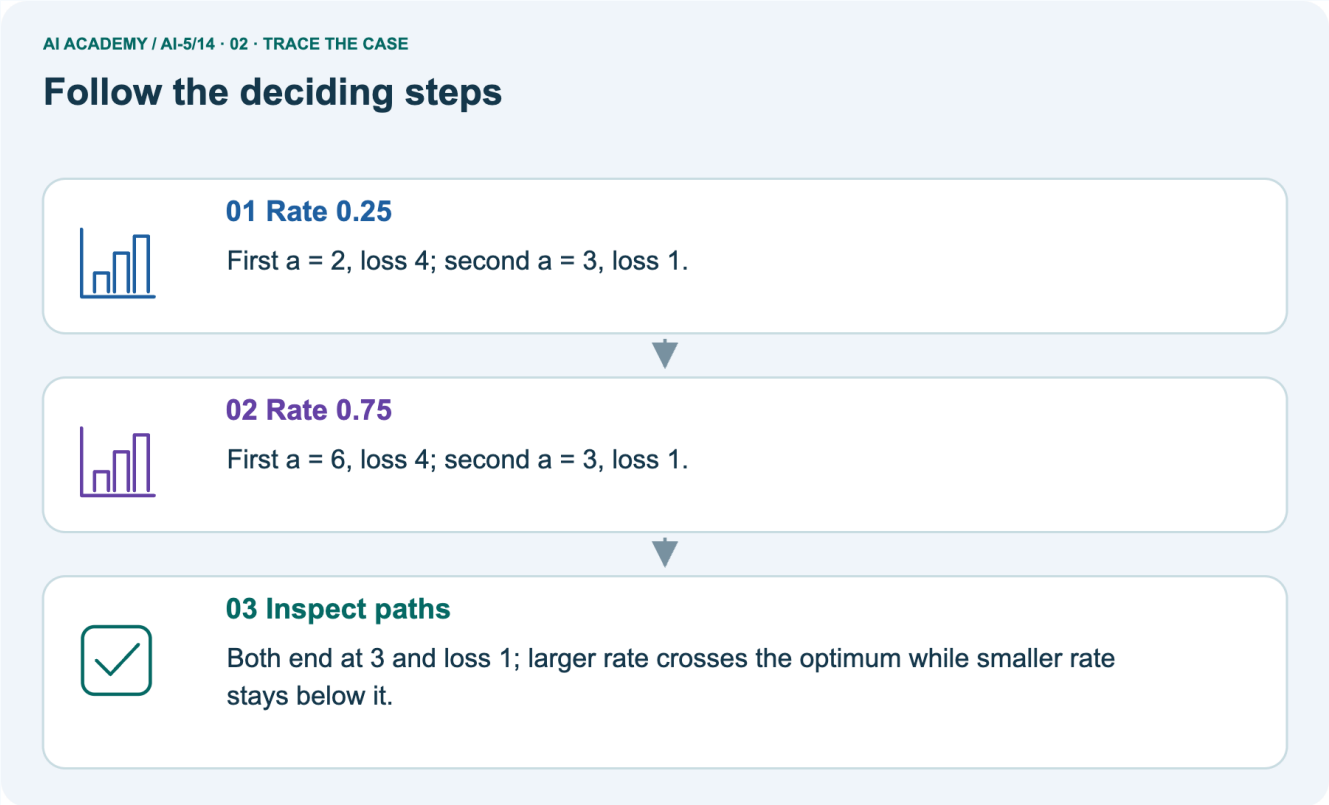
Compare **learning rate**s with identical starting loss and update counts. Supplied facts: Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



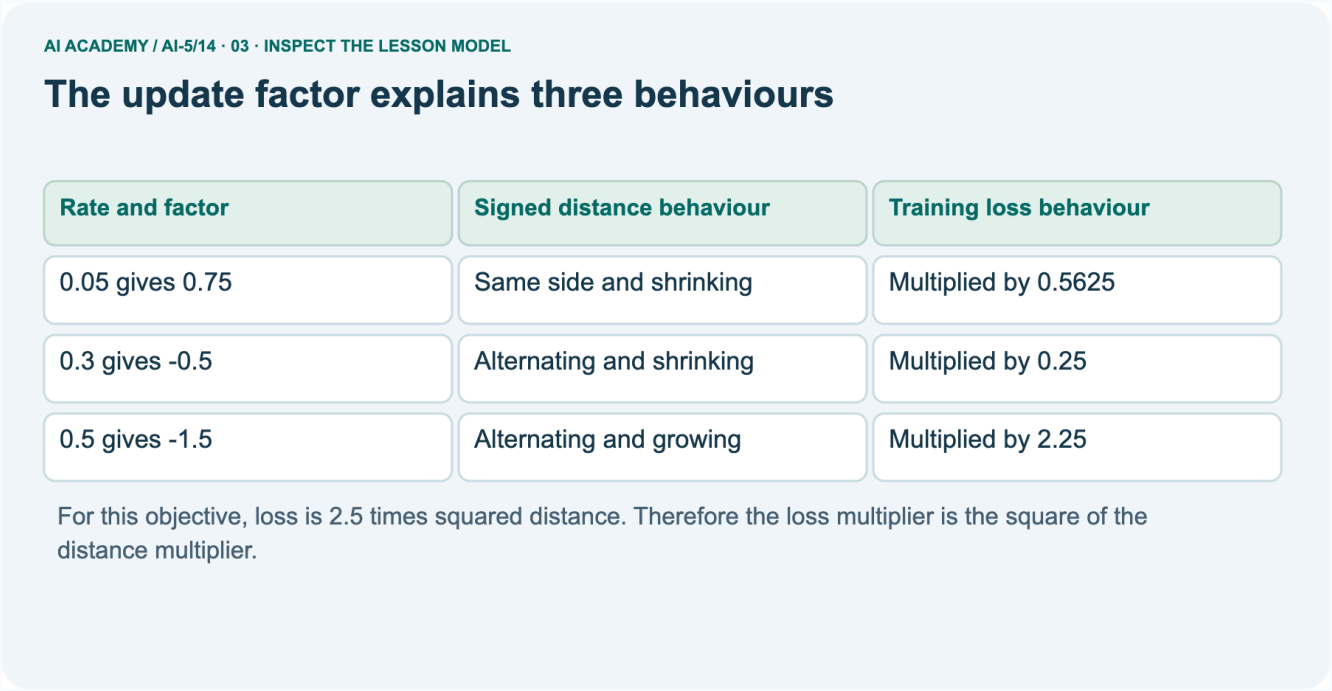
Follow the labelled picture with the matching explanation. Open the vector diagram

Rate 0.25: First a = 2, loss 4; second a = 3, loss 1. → Rate 0.75: First a = 6, loss 4; second a = 3, loss 1. → Inspect paths: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The update factor explains three behaviours



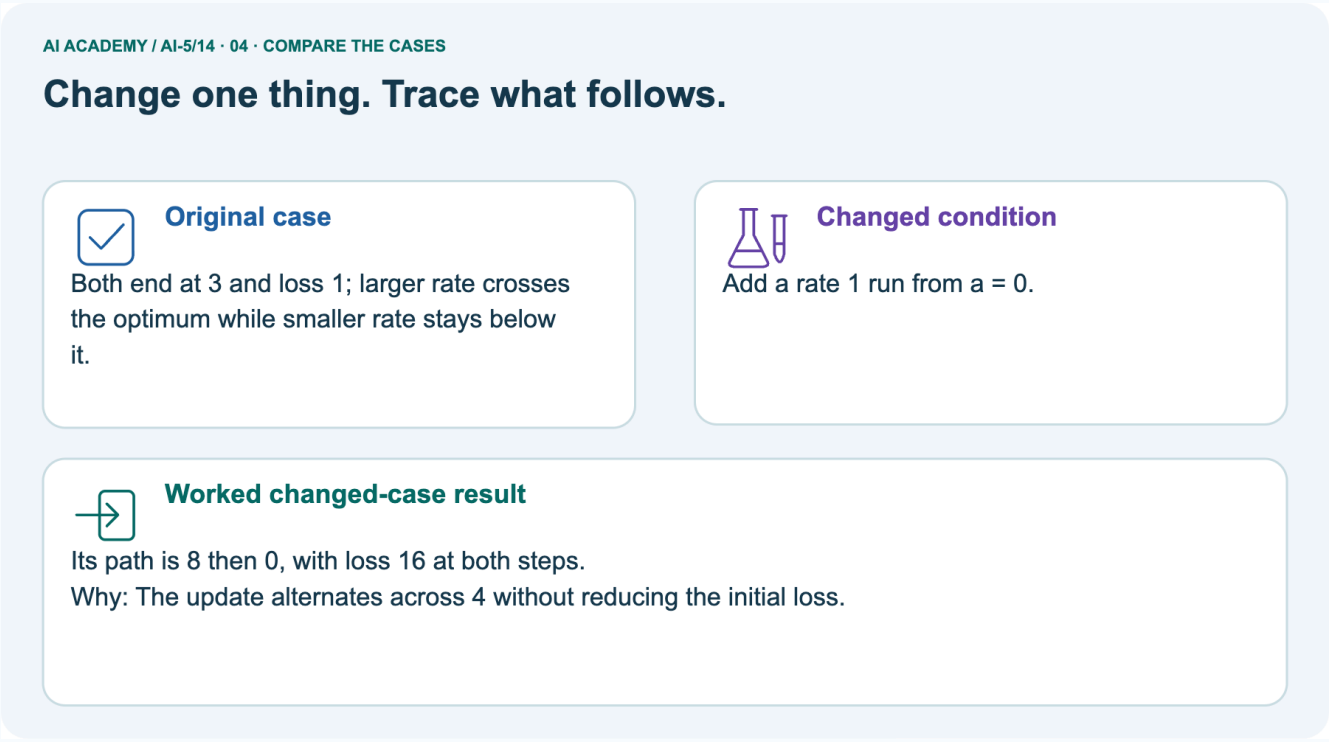
Follow the labelled picture with the matching explanation. Open the vector diagram

The update factor explains three behaviours. 0.05 gives 0.75 | Same side and shrinking | Multiplied by 0.5625. 0.3 gives -0.5 | Alternating and shrinking | Multiplied by 0.25. 0.5 gives -1.5 | Alternating and growing | Multiplied by 2.25

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

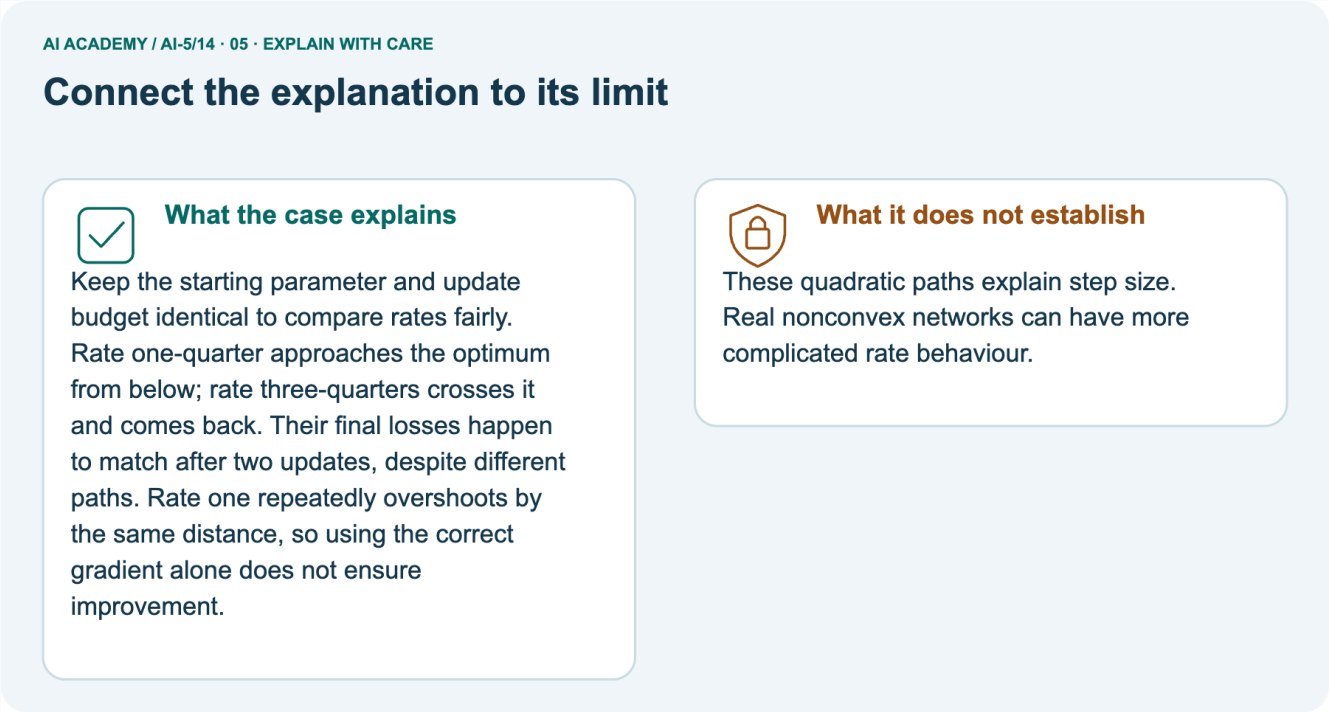
Original case: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Changed condition: Add a rate 1 run from a = 0. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

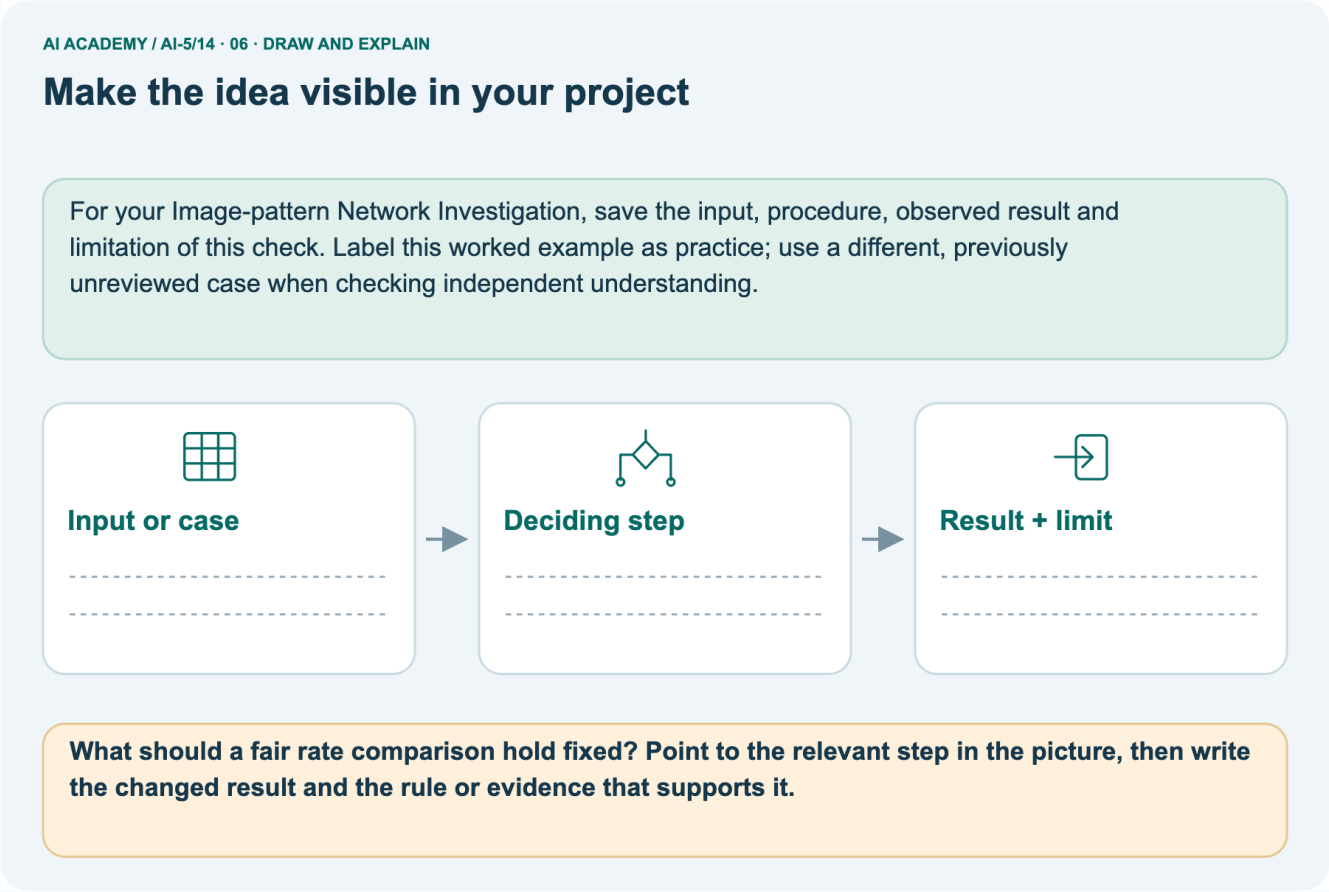
Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Limit: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Learning rate experiments: the case

02. Follow the deciding steps

03. The update factor explains three behaviours

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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